SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 2.1: Similarity and Enlargement — Teacher Reference (Pre-filled)

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 2.1: Similarity and Enlargement
Driving Question	DRIVING QUESTION: Why does doubling the size of a container make it hold eight times as much — and how can we use this mathematical pattern to design, build, and predict the properties of any similar shape or object?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **What Makes Two Shapes 'the Same but Different'? — Introducing Similarity**	Students examined pairs of real objects (a small and large Uji cup, a passport photo and an enlarged print, a Kenyan shilling coin next to a scaled drawing) and measured corresponding sides and angles. They recorded that angles stayed equal while side lengths changed by a consistent ratio.	Two figures are similar if all corresponding angles are equal AND all corresponding sides are in the same ratio (the linear scale factor k). Congruent figures are a special case where k = 1. Teacher note: Stress that BOTH conditions must hold simultaneously — equal angles alone (as in a rectangle vs. a non-square rectangle) are not sufficient.	Students explained why the passport photo and its enlargement look identical in shape but differ in size: every length was multiplied by the same k, preserving all angles. They linked this to the driving question by noting that if k controls lengths, something more dramatic must happen to area and volume — setting up the inquiry for Lessons 4–5. Pedagogical note: Cold-call at least three students to state the definition of similarity in their own words before moving on.
Lesson 2 **Centre of Enlargement and Linear Scale Factor**	Students drew enlargements of a triangle on grid paper using a given centre of enlargement O and scale factors k = 2 and k = 3. They measured the distances OA, OA′, OB, OB′, OC, OC′ and verified that OA′/OA = OB′/OB = OC′/OC = k. They also observed that O, each object vertex, and its image vertex are always collinear.	The centre of enlargement O is the unique fixed point from which every vertex of the object is projected to its image vertex such that the image distance from O equals k × the object distance from O. A negative scale factor maps the image to the opposite side of O. Teacher note: Require students to draw ray lines through O and each vertex — this geometric habit reinforces the definition and catches arithmetic errors.	Students explained how an architect in Nairobi uses a centre of projection (analogous to O) when producing scaled blueprints: every measurement on the plan is exactly k times the real dimension, all radiating from a reference origin. They revised their similarity definition to include the role of O, updating their DQB with the question 'Does changing O change the shape or size of the image?'. Pedagogical note: After the DQB update, allow 30 seconds of wait time before cold-calling two students to answer that question.

Lesson 3 **Finding the Centre of Enlargement and Scale Factor from Object and Image**	Given pre-drawn object and image pairs (without O marked), students used a ruler to draw lines through each pair of corresponding vertices and found that all lines met at a single point — the centre of enlargement O. They then calculated k by dividing one image side by the corresponding object side, confirming consistency across all sides.	To find O, draw straight lines through at least two pairs of corresponding vertices; their intersection is O. The linear scale factor k = (image length) ÷ (object length) for any corresponding pair of sides. If $k > 1$ the image is an enlargement; if $0 < k < 1$ it is a reduction; if $k < 0$ the image is on the opposite side of O. Teacher note: Insist students verify k using a second pair of sides — this builds the habit of checking consistency that is essential for problem-solving questions.	Students explained how a cartographer at the Kenya National Bureau of Statistics locates the 'projection centre' when scaling a district map, and why every road length on the map must satisfy the same ratio k to its real-world length. They connected this reverse-engineering skill (finding O and k from given figures) to the driving question: if we can recover k from any two similar objects, we can predict lengths, and — as upcoming lessons will show — areas and volumes too. Pedagogical note: Use a think-pair-share (1 min pair, cold-call three pairs) to consolidate the method before the explain phase.
Lesson 4 **Area Scale Factor: Why Does the Uji Cup Label Need Nine Times as Much Paper?**	Students wrapped rectangular paper labels around small ($k = 1$) and large ($k = 3$) cylindrical Uji cups, then cut the labels flat and counted grid squares to compare areas. They recorded that the large label covered 9 times as many squares as the small one, even though the cup was only 3 times taller and 3 times wider.	The area scale factor of similar figures equals k^2, because both the length and the width of any region are each multiplied by k, making the total area multiply by $k \times k = k^2$. Therefore: if $k = 3$, area scale factor = 9; if $k = 2$, area scale factor = 4. Teacher note: A common misconception is that area scales by k. Address this directly by asking: 'If I double every side of a square, how many small squares fit inside the big one?' — the 2×2 grid shows 4, not 2.	Students explained the label problem: the printer in Mombasa packaging factory must order $k^2 = 9$ times as much paper for the large cup label as for the small cup label, even though $k = 3$ feels like 'three times as big.' They generalised: any painter, tiler, or fabric cutter working with similar shapes must multiply material costs by k^2, not k. This partial answer to the driving question (area $\propto$ k^2) was posted on the DQB alongside the still-open volume question. Pedagogical note: Require students to write the formula $Area_{image} = k^2 \times Area_{object}$ in their notes before the explain discussion.
Lesson 5 **Volume Scale Factor: Why the Large Uji Cup Holds Eight Times as Much**	Students filled a small cubic container (side 1 unit) with water and poured it into a large similar container (side 2 units, $k = 2$), counting how many small-cube fills were needed. They repeated with $k = 3$ using small and large rectangular boxes. Results: $k = 2$ required 8 small fills; $k = 3$ required 27 small fills.	The volume scale factor of similar solids equals k^3, because length, width, and height are each multiplied by k, so volume multiplies by $k \times k \times k = k^3$. For $k = 2$: volume scale factor = 8; for $k = 3$: volume scale factor = 27. Teacher note: This is the direct answer to the driving question's anchor phenomenon. Make the connection explicit: 'Doubling every dimension gives $k = 2$, so volume scales by $2^3 = 8$ — that is precisely why the big container holds eight times as much.'	Students explained how a water-tank manufacturer in Nakuru deciding between a 1 000-litre and a 2 000-litre tank cannot simply double the dimensions — doubling dimensions would produce an 8 000-litre tank ($k^3 = 8$). They calculated the correct k for a 2 000-litre tank: $k = \sqrt[3]{2} \approx 1.26$, meaning each dimension increases by only about 26%. This counter-intuitive result directly answers the driving question and was recorded as the key insight on the DQB. Pedagogical note: Cold-call four students to state the k^3 rule in context before closing the lesson.
Lesson 6 **Relating Linear, Area**	Students completed a structured table: given k, they computed k^2, k^3, and verified	For any two similar figures or solids with linear scale factor k: corresponding lengths	Students explained a unified design problem: a Kisumu ceramics studio scales

and Volume Scale Factors — The Unified k, k^2, k^3 Model	each against a physical or drawn model. They also worked backwards — given area scale factor or volume scale factor, they recovered k. Examples used similar triangles cut from paper (area) and similar cuboids built from unit cubes (volume) in a Kenyan packaging context (maize flour bags of different sizes).	scale by k, corresponding surface areas (and any area measurement) scale by k^2, and corresponding volumes scale by k^3. Conversely: $k = \sqrt{\text{area scale factor}} = \sqrt[3]{\text{volume scale factor}}$. Teacher note: The table format ($k \mid k^2 \mid k^3$) is a powerful reference tool — have students keep a completed version in their exercise books as a personal formula card for examinations.	up a small decorative pot (height 8 cm, surface area 150 cm^2, volume 200 cm^3) by k = 2.5. They predicted: new height = 20 cm, new surface area = 150 × 6.25 = 937.5 cm^2, new volume = 200 × 15.625 = 3 125 cm^3. They confirmed that knowing only k is sufficient to scale every measurable property of a similar object — directly answering the second part of the driving question ('predict the properties of any similar shape'). Pedagogical note: Assign one 'backwards' problem (given volume ratio, find surface area ratio) as an exit ticket.
Lesson 7 **Real-Life Applications and Problem Solving — Apply & Communicate**	Students worked through four contextualised problem sets: (1) map-scale problems using a Nairobi city map (area scale factor for a park); (2) recipe scaling — a githeri recipe for 4 people versus 32 people using similar-volume pots; (3) a Kenyan athlete's water bottle (k = 1.5 scaling to a team jug); (4) a construction problem comparing the cost of painting two similar water towers near Eldoret. Students identified which scale factor (k, k^2, or k^3) applied in each context.	The choice of scale factor depends on what is being measured: use k for lengths and distances (map scales, heights, widths); use k^2 for areas (paint coverage, fabric, labels, map areas, land parcels); use k^3 for volumes (liquid capacity, mass of similar objects of the same material, ingredient quantities). Teacher note: The most common exam error is applying k where k^2 or k^3 is needed. Explicitly name this 'the scale-factor selection error' and give students a decision flowchart: 'Is it a length? → k. An area? → k^2. A volume? → k^3.'	Students explained each solution to a partner, then selected one problem to present to the class with a written justification for their choice of k, k^2, or k^3. They revised their DQB to mark which driving-question components were now fully answered and noted remaining questions (e.g., 'What if the shapes are similar but not the same material?'). Pedagogical note: Require every student to write a one-sentence 'because' statement justifying their scale-factor choice — this surfaces lingering misconceptions before the final lesson.
Lesson 8 **Project Presentations, Final Explanation & DQB Completion**	Student groups presented their similarity-and-enlargement projects (e.g., scaled models of Kenyan buildings, packaging redesigns, map-distance analyses). Each group displayed their object and image measurements, calculated k, k^2, and k^3, and demonstrated at least one prediction verified against a physical measurement or calculation.	Consolidated understanding: (1) similar figures have equal corresponding angles and proportional corresponding sides; (2) the linear scale factor k is the constant ratio of any image length to its object length; (3) area scales by k^2 and volume scales by k^3 — these three relationships form a unified, predictive model applicable to any similar shape or solid. Teacher note: Use the completed DQB as a whole-class review tool — read each original question aloud, ask students to vote (thumbs up/down) on whether it is now answered, and cold-call students to provide the answer. This meta-cognitive closure is essential for retention.	Students gave a final written response to the driving question: 'Doubling a container's dimensions means k = 2; volume scales by k^3 = 8, so it holds eight times as much. We can use the k, k^2, k^3 model to design and predict the length, surface area, and volume of any similar shape by knowing just one scale factor.' Pedagogical note: Collect written responses as a summative assessment. Check for three elements: correct identification of k, correct application of k^2 for area, and correct application of k^3 for volume, each in a real-world Kenyan context.